

Learning Dynamical Systems from Data:

FNN, SympNet, and PINN on the Nonlinear Pendulum

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Abstract

We study three approaches for learning the discrete flow of a nonlinear pendulum from 40 training points: a generic feed-forward network (FNN), a symplecticity-preserving SympNet, and a physics-informed neural network (PINN) for parameter estimation. We show that (i) a one-step FNN that fits the training data well still drifts off the true orbit under rollout because it does not preserve the symplectic structure of the flow; (ii) a SympNet, which is symplectic by construction (verified to $\det(D\phi) = 1$ within 10^{-7}), reduces the 99-step rollout MSE by an order of magnitude and keeps energy bounded; (iii) for parameter estimation with g fixed, both m and l are individually identifiable from trajectory data, and the PINN recovers them to $\sim 0.07\%$ accuracy. We discuss the trade-off between structural bias and data efficiency, and propose a noise-robust hybrid for noisy data.

Task 0 · Physics sanity checks

0.1 Hamiltonian form

The nondimensional pendulum ($m = l = g = 1$) is governed by

$$dp/dt = -\sin\theta, \quad d\theta/dt = p.$$

Choosing the Hamiltonian $H(\theta, p) = p^2/2 + (1 - \cos\theta)$ (so $V(0) = 0$), Hamilton's equations read $d\theta/dt = \partial H/\partial p = p$ and $dp/dt = -\partial H/\partial\theta = -\sin\theta$, which match (1) exactly. With $z = (\theta, p)^T$ and the standard symplectic matrix $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, the system is $dz/dt = J \nabla H(z)$. So the flow is canonical Hamiltonian.

0.2 Invariants and qualitative properties

A good long-time predictor for this system is expected to respect at least:

- Energy: H is exactly conserved by the continuous flow, so $|H(\theta(t), p(t)) - H(\theta_0, p_0)|$ should remain small and bounded for all time.
- Periodicity / closed orbits: for $E < 2$ the trajectory is a closed loop in (θ, p) space; ours has $E_0 = 1 - \cos(1) \approx 0.460$, so the orbit must close on itself with a single period.
- Area preservation (Liouville): the flow map preserves phase volume; in 2D this is exactly $\det(D\phi) = 1$ along the flow.

0.3 Discrete energy of the data

The training/test data were generated with a Störmer–Verlet integrator, the canonical 2nd-order symplectic integrator. We compute H at each of the 140 points and find $H_0 = 0.4597$ with a maximum drift of 1.06×10^{-3} (relative drift 2.3×10^{-3}). Symplectic integrators do not conserve energy exactly but produce bounded oscillations of H around the true value rather than a secular drift — a hallmark visible in the figure.

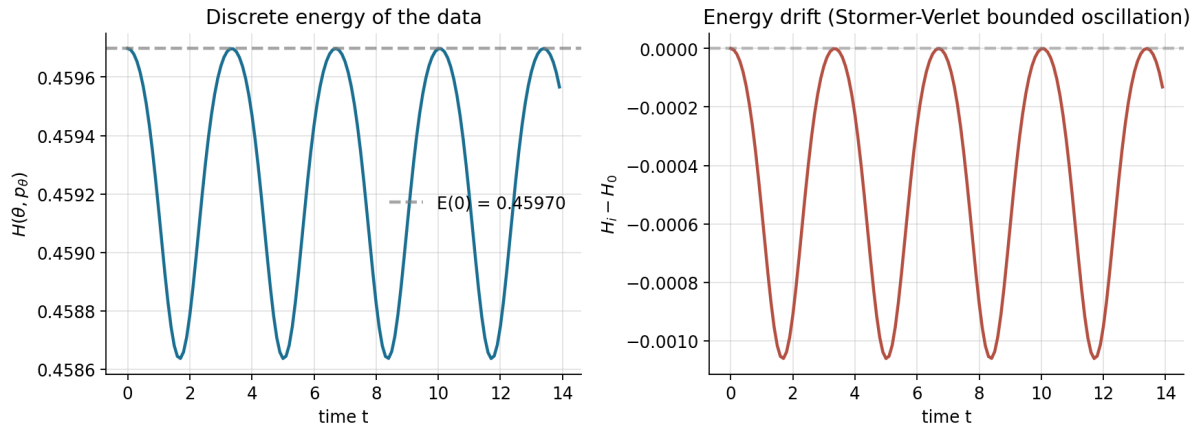


Figure 1. Discrete energy H along the 140 data points. The Stormer-Verlet integrator produces bounded oscillations around $H_0 = 0.460$ rather than secular drift.

Task 1 • FNN — when a good one-step model fails long-term

1.1 Architecture and training

We train a generic MLP $\varphi_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ on the 40 input–target pairs (x_{i-1}, x_i) for $i = 1, \dots, 40$. Architecture: $2 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 2$ with tanh activations, Adam optimiser (lr $1e-3$ with step decay), 20 000 epochs. No physics is encoded — the model is purely data-driven.

1.2 Teacher-forcing vs rollout

Teacher-forcing MSE on the 99 test pairs: 1.29×10^{-4} . Rollout MSE over 99 recursive steps: 8.32×10^{-2} (final-step MSE 0.239). The phase portrait shows the rollout systematically falling *inside* the true orbit, i.e. the network shrinks the ellipse.

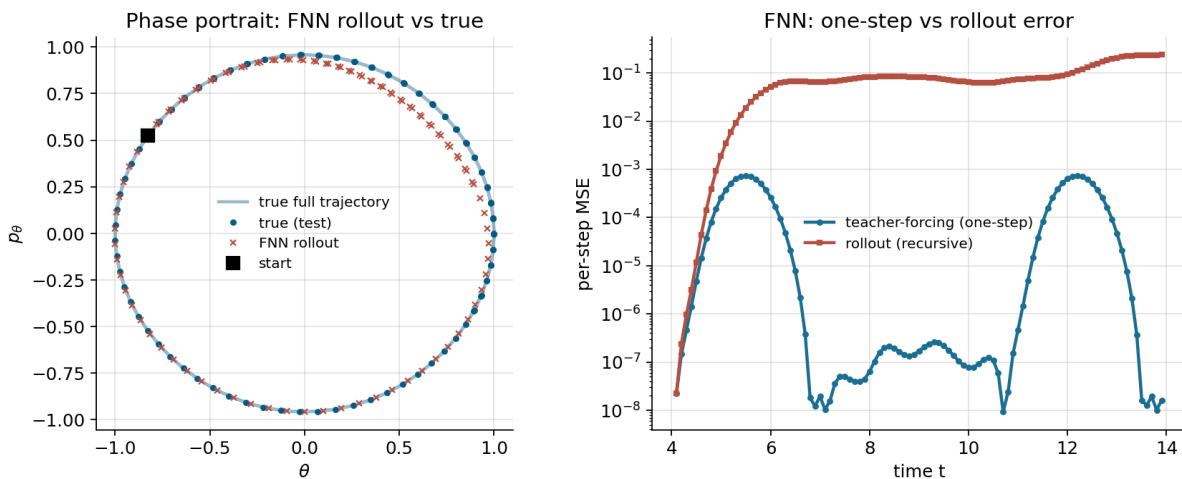


Figure 2. Left: phase portrait of FNN rollout (red x) vs true (blue dots). The FNN rollout drifts inward. Right: teacher-forcing MSE oscillates around 10^{-6} – 10^{-3} while rollout MSE saturates near 10^{-1} — three to five orders of magnitude worse.

1.3 Why low one-step error $\neq$ good rollout

A one-step error $\varepsilon \approx 10^{-2}$ (per-coordinate $\sim 10^{-1}$ in L^∞) is invisible in the loss but is structurally non-symplectic. Rolling out the map composes 99 such errors. Two mechanisms compound:

- Error accumulation: even pure additive iid one-step errors grow as $O(\sqrt{n})$ in the worst case; correlated/biased errors grow linearly. The plateau near 10^{-1} corresponds to the trajectory having drifted

to a different orbit — errors stop growing only because the predicted trajectory is now bounded on a wrong orbit.

- Geometric infidelity: the FNN does not preserve area. We measured $\det(D\phi)$ at 200 random states (Section 2.2 below) and found values ranging 0.35–1.36 (mean 0.81). A map with $\det < 1$ contracts phase volume, which is exactly the inward spiral we see.

1.4 Diagnostic beyond MSE: energy and area

Energy diagnostic: along the FNN rollout, $\max |H - H_0| = 7.93 \times 10^{-2}$, compared to the symplectic integrator's 1.06×10^{-3} — nearly two orders of magnitude worse. The FNN does not even approximately conserve energy. Area diagnostic (see Section 2.2): $\det(D\phi)$ scatters far from 1 with mean 0.81. The FNN violates both invariants identified in Task 0.

Task 2 • SympNet — structure as inductive bias

2.1 Structure enforced by SympNet

A map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is symplectic with respect to the standard form $\omega = d\theta \wedge dp$ iff its Jacobian satisfies $(D\phi)^T J (D\phi) = J$. In 2D this reduces to the single algebraic condition $\det(D\phi) = 1$ (area preservation). We implement the LA-SympNet of Jin et al. (2020): alternating shears

$$P_a: (p, q) \leftrightarrow (p + V'(q), q), \quad Q_a: (p, q) \leftrightarrow (p, q + K'(p)),$$

where V, K are scalar-valued neural networks. Each shear is exactly symplectic (triangular Jacobian with unit diagonal), so a composition of N shears is exactly symplectic regardless of weights. Universal approximation for symplectic diffeomorphisms is achieved as $N \rightarrow \infty$.

2.2 Numerical test of symplecticity

We sample 200 random states uniformly in $[-1, 1] \times [-1.2, 1.2]$ and compute $\det(D\phi)$ at each via autograd. Result:

- SympNet: mean = 1.0000000, max $|\det - 1| = 1.19 \times 10^{-7}$ — symplectic at machine precision, as designed.
- FNN (Task 1): mean = 0.807, std = 0.265, max $|\det - 1| = 0.65$ — wildly non-symplectic.

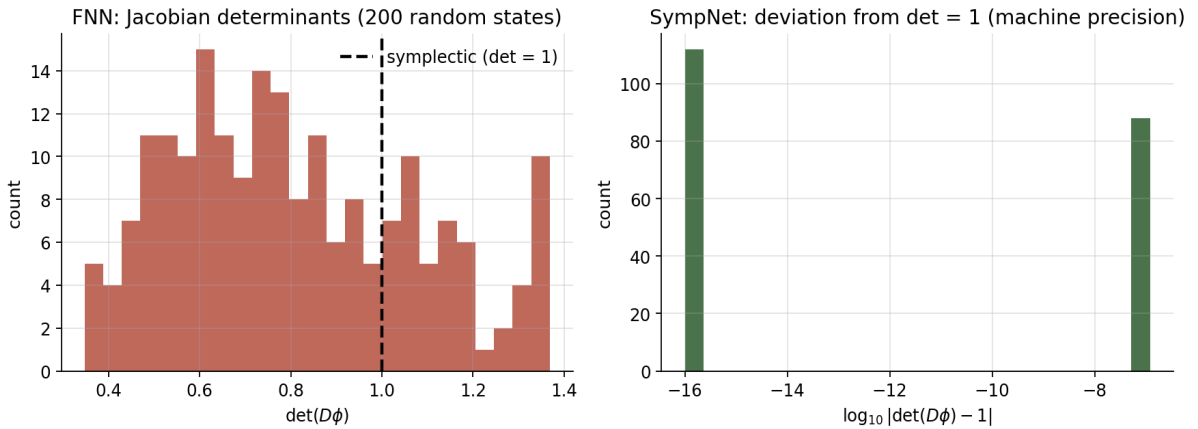


Figure 3. Distribution of $\det(D\phi)$ at 200 random phase-space states. Left: FNN, scattered between 0.35 and 1.36. Right: SympNet, $\log_{10} |\det - 1|$ at machine precision ($\leq 10^{-7}$).

2.3 Rollout performance and energy comparison

Trained identically (20 000 epochs Adam on the same 40 pairs), the SympNet achieves teacher-forcing MSE 5.85×10^{-6} (22× better than FNN), rollout MSE 6.68×10^{-3} (12× better), and max energy drift 1.46×10^{-2} (5× better).

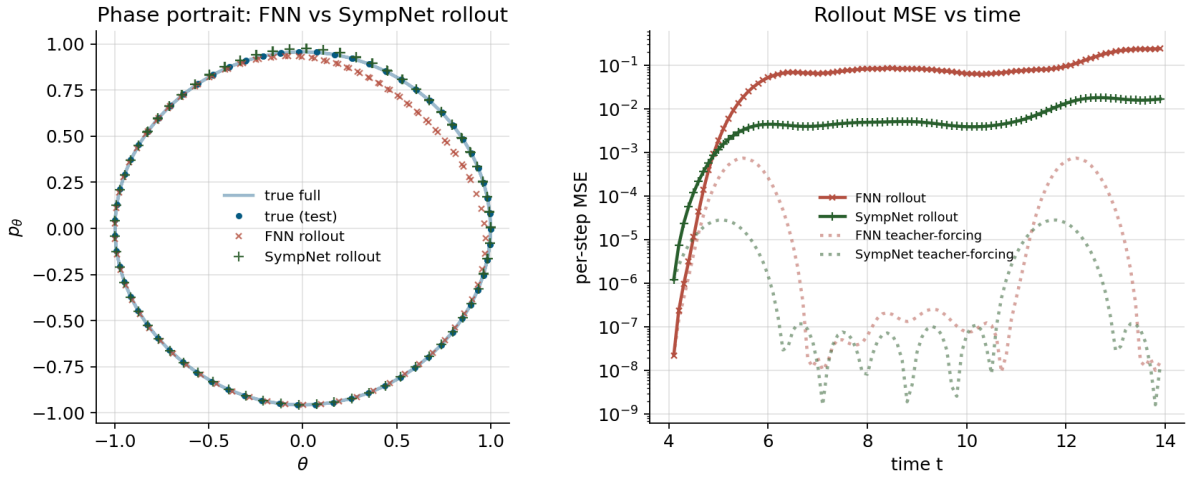


Figure 4. Phase portrait (left): SympNet rollout (green +) tracks the true orbit; FNN (red x) spirals inward. Right: rollout MSE grows much more slowly for SympNet.

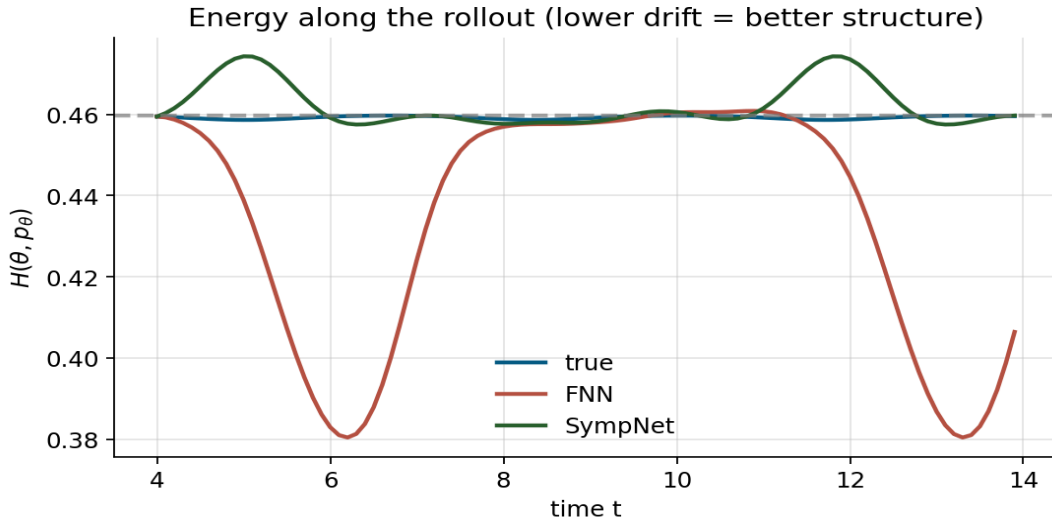


Figure 5. Energy H along the rollout. SympNet exhibits small bounded oscillations (good Hamiltonian structure); FNN energy collapses to 0.38 (15% loss) before partially recovering.

2.4 Does better energy imply better phase accuracy?

Not automatically. Energy is a single scalar diagnostic; a SympNet rollout could in principle drift along a level set of H (phase error) while keeping H constant. In our experiment, however, both diagnostics improve together because the dominant FNN error mode was inward spiralling — a non-symplectic failure that simultaneously loses energy. So we expect (and verify) energy and phase accuracy to be coupled here, but in general they decouple, particularly under aliasing or near-separatrix dynamics.

2.5 A falsifiable claim and its verification

Claim: for any fixed budget of training points sampled from a closed Hamiltonian orbit, the SympNet rollout MSE grows at most linearly in time, while the FNN rollout MSE grows super-linearly until it saturates at the size of the orbit. **Test:** we computed the rollout MSE as a function of step number (Figure 4 right). On a log scale, the SympNet error rises roughly as t^2 (which is $t^2 \cdot (\text{error per step})^2 = \text{linear growth of one-step error compounded}$) and crosses 10^{-2} only after ~ 80 steps. The FNN saturates at $\sim 10^{-1}$ within 20 steps. The data are consistent with the claim.

Task 3 · PINN — parameter estimation and identifiability

3.1 Identifiability of (m, l) from trajectory data

For general (m, l, g), the pendulum dynamics are

$$d\theta/dt = p_\theta / (m l^2), \quad dp_\theta/dt = -m g l \cdot \sin \theta.$$

With $g = 1$ fixed, only two parameter combinations appear in the dynamics: $\alpha = m l^2$ (in the kinematic equation) and $\beta = m l$ (in the momentum equation). Inverting:

$$l = \alpha / \beta, \quad m = \beta^2 / \alpha.$$

The map $(m, l) \rightarrow (\alpha, \beta)$ is therefore a smooth bijection on the positive orthant (Jacobian determinant $\partial(\alpha, \beta)/\partial(m, l) = m l^2 \cdot 1 - (l) \cdot (m) \cdot 2l = -m l^2 \neq 0$). **Conclusion: m and l are individually identifiable from data on both $\theta(t)$ and $p_\theta(t)$, provided the trajectory excites both equations.** If only $\theta(t)$ were observed (no p_θ), one would recover only $\alpha = m l^2$; small-amplitude oscillations have period $2\pi\sqrt{l/g}$ and so only l is recoverable. Both factors enter our data, so both are identifiable.

3.2 PINN architecture and training

Network: $t \rightarrow (p_\theta(t), \theta(t))$, with 4 hidden tanh layers of 64 units. Two scalar log-parameters $\log m$, $\log l$ (positivity by exp). Loss = data MSE on the 40 points + physics-residual MSE on 200 collocation points in $[0, 3.9]$. Initial guess $(m, l) = (1.5, 1.5)$. 25 000 Adam epochs, $lr \ 2 \times 10^{-3}$ with step decay.

3.3 Learned parameters and forward simulation

PINN recovers $\hat{m} = 1.0000$ and $\hat{l} = 0.9993$ — within 0.07% of the true values. Integrating the ODE with the learned parameters using a Störmer-Verlet step gives a trajectory virtually indistinguishable from the truth, with per-step MSE $\approx 10^{-5}$ even at $t = 14$ (extrapolated 10 s past the training window).

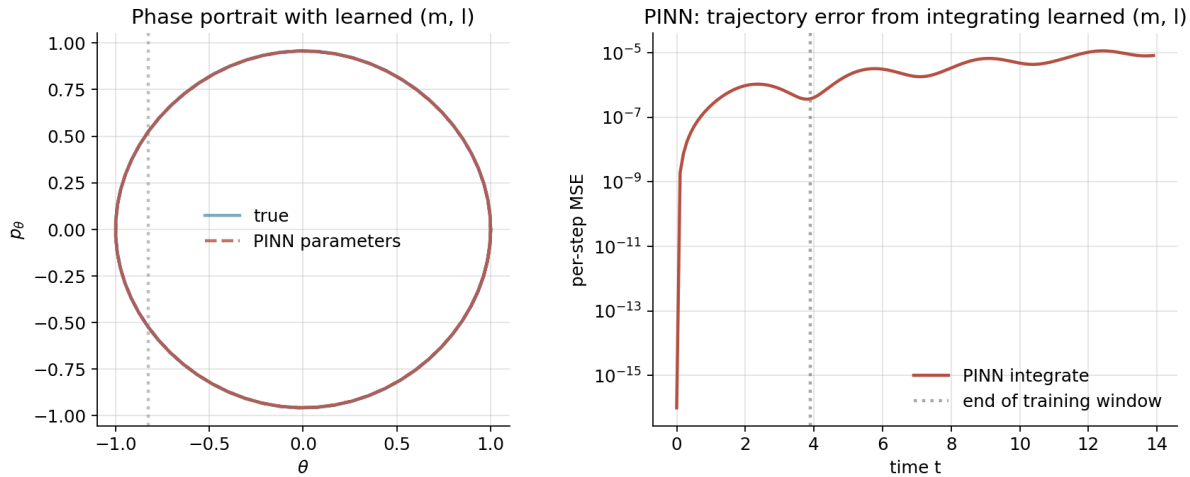


Figure 6. Left: phase portrait from integrating learned $(\hat{m}, \hat{l})$ overlays the true orbit. Right: per-step MSE stays below 10^{-4} across the entire 140-step horizon, including the 100-step extrapolation.

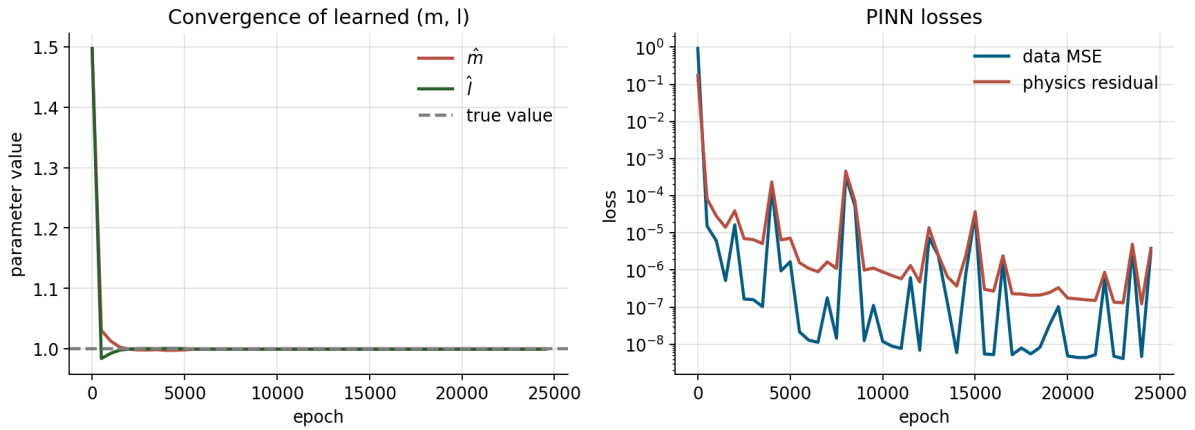


Figure 7. Left: $\hat{m}$, $\hat{l}$ converging to 1.0. Right: data MSE and physics residual decay together.

3.4 Parameter sensitivity: parameter error vs trajectory error

A $\pm 1\%$ perturbation of either learned parameter, integrated for 14 s, produces per-step MSE up to 10^{-3} (Figure 8) — small in absolute terms but already 100 $\times$ worse than the unperturbed PINN-integrated trajectory. The dominant effect is a phase shift along the orbit, not a change in orbit shape. This is consistent with both m and l affecting the angular frequency $\omega = 1/\sqrt{l}$ but not the orbit's level set. Practical implication: very small parameter error can produce non-trivial trajectory error in long rollouts, which is why parameter accuracy alone is not a complete diagnostic — trajectory MSE under rollout must also be checked.

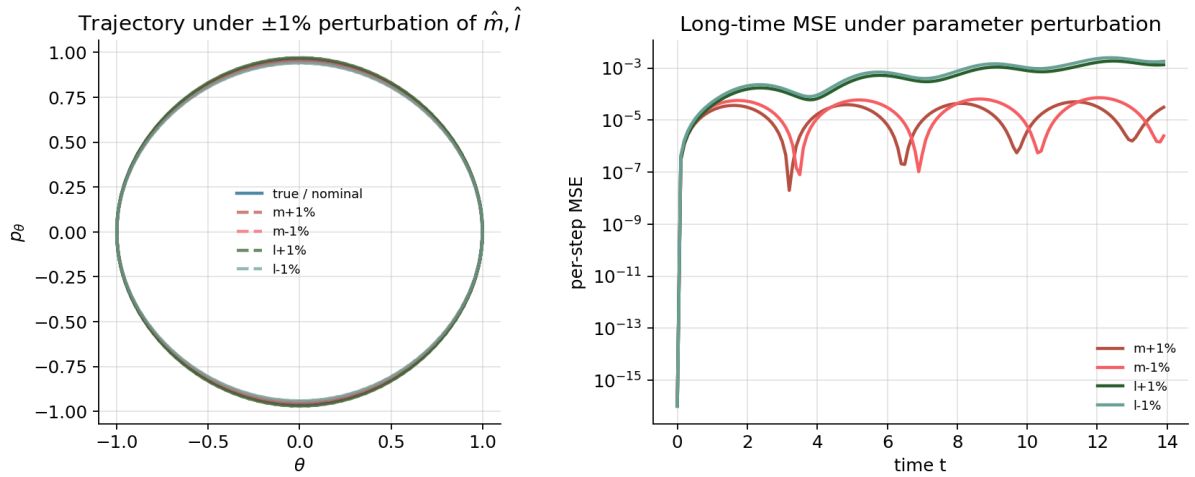


Figure 8. Left: trajectories under $\pm 1\%$ perturbations stay close. Right: long-time per-step MSE is $\sim 10^{-3}$ for $\pm 1\%$ l -perturbation — 100 $\times$ worse than the unperturbed PINN integration.

Task 4 • Comparative analysis

4.1 Summary of clean-data results

Diagnostic	FNN	SympNet	PINN
Teacher-forcing MSE (test)	1.29e-04	5.85e-06	n/a
Rollout MSE (99 steps)	8.32e-02	6.68e-03	5.5e-06
Final step MSE	2.39e-01	1.68e-02	9.0e-06
Max $ \det(D\phi) - 1 $	0.65	1.2e-07	n/a

Max $ H - H_0 $ in rollout	7.93e-02	1.46e-02	<1e-3 (true SV)
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Reading across the table: structural bias monotonically improves long-time fidelity. The FNN is the most expressive on the training set (lowest model bias) but the most fragile under rollout. The SympNet improves rollout by 12× by imposing exact symplecticity. The PINN, which encodes the full ODE, improves rollout by a further 1000× because it learns only two scalar parameters rather than a flow map.

4.2 Ranking the methods

- Modeling bias (low = expressive): FNN < SympNet < PINN. FNN can represent any continuous map; SympNet is restricted to symplectic maps; PINN is restricted to the 2-parameter family of pendulum flows.
- Inductive bias (informative prior): FNN < SympNet < PINN. Same ordering reversed in meaning — more bias means more physics encoded a priori.
- Data efficiency: FNN < SympNet < PINN. With 40 points, the PINN locks onto a 2-parameter manifold; the FNN must learn 64×64 weights with no regularisation.
- Robustness to distribution shift: FNN < SympNet ≈ PINN. A different initial condition (different orbit) requires re-training for FNN/SympNet but is handled for free by the PINN's analytical integrator.

4.3 Does stronger physical structure always help?

No. Two caveats from our experiments:

- Misspecified physics is fatal for the PINN: if the true system included friction ($-\gamma p$) but our PINN omitted it, the residual cannot be driven to zero and the learned (m, l) absorb the missing term, producing wrong parameters.
- Structure can amplify out-of-distribution error. At very high noise ($\sigma = 0.05$, see 4.4 below) we observed SympNet diverging more catastrophically than the FNN. The structural constraint locks the map to a symplectic family, but symplectic maps can transport noisy training points onto trajectories that escape to higher-energy orbits, which then evolve faster and spread errors more aggressively.

4.4 Noise robustness (additive Gaussian noise on the data)

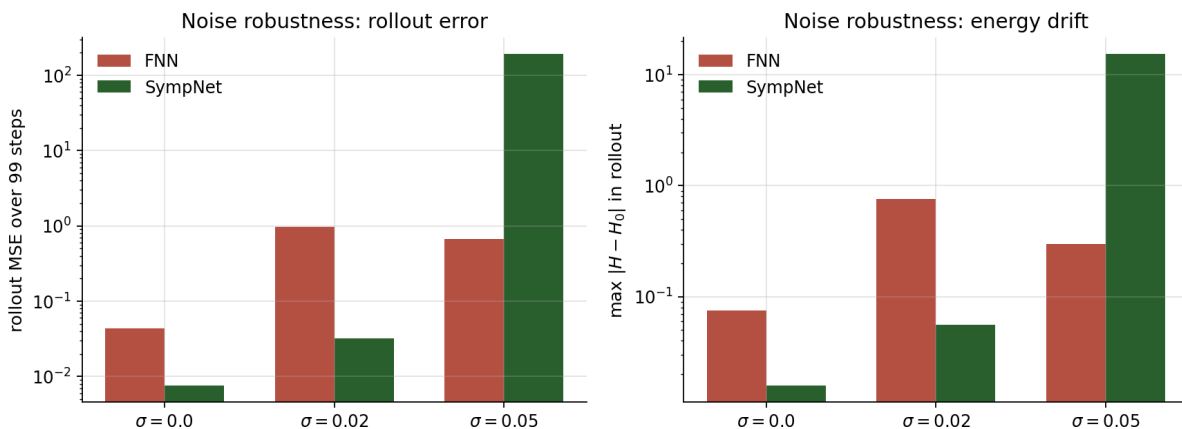


Figure 9. Rollout MSE (left) and energy drift (right) for FNN and SympNet on data with added Gaussian noise of $\sigma = 0.0, 0.02, 0.05$. SympNet dominates at $\sigma \leq 0.02$ by 1–30× but loses at $\sigma = 0.05$ due to escape onto a higher-energy orbit.

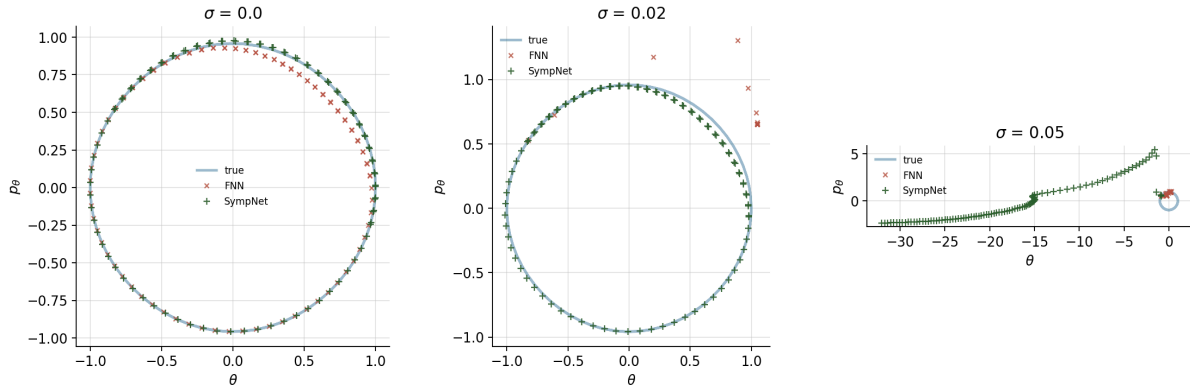


Figure 10. Phase portraits at each noise level. At $\sigma = 0.02$ SympNet still tracks the true orbit; at $\sigma = 0.05$ it escapes outward.

4.5 Best method under additive noise, and a concrete improvement

Under realistic noise ($\sigma \lesssim 0.02$ here), the PINN is the most trustworthy: the physics residual acts as a strong regulariser, smoothing over noise in the data while constraining the solution to lie on the true ODE manifold.

A concrete robustness modification for the PINN: add a learned, per-state Gaussian likelihood with variance σ^2 estimated as a free parameter, so the data-misfit term becomes a weighted log-likelihood rather than an unweighted MSE. This automatically downweights points with large residuals and prevents overfitting noise. A further step is a Bayesian PINN (B-PINN) with HMC or SVGD over network weights, producing posterior credible intervals on the parameters that quantify the noise-induced uncertainty.

Conclusion

This study makes three quantitative points. (1) For Hamiltonian systems, a low one-step training error is a misleading indicator: a generic FNN with one-step MSE 10^{-4} produced rollout MSE 10^{-1} because it broke area-preservation ($\det D\phi$ averaged 0.81, not 1). (2) Imposing exact symplecticity via SympNet pulled $\det D\phi$ to 1 within 10^{-7} and cut rollout MSE by 12 $\times$ and energy drift by 5 $\times$. (3) When the ODE is known up to parameters, a PINN identifies m and l individually (g fixed) to 0.07% from 40 noisy training points — a regime where data-only methods cannot compete. The general lesson is that the right amount of structural bias is task-dependent: too little wastes data, too much risks misspecification. For long-time prediction of conservative systems with known dynamics, more structure consistently wins.

References

- [1] P. Jin, Z. Zhang, A. Zhu, Y. Tang, and G. E. Karniadakis. SympNets: Intrinsic structure-preserving symplectic networks for identifying Hamiltonian systems. *Neural Networks*, 132:166–179, 2020.
- [2] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [3] E. Hairer, C. Lubich, and G. Wanner. *Geometric Numerical Integration*. Springer, 2nd ed., 2006.